Docker & WSL Troubleshooting Log

Comprehensive sequence resolving permissions, virtualization setups, and kernel layers on Windows

Environment: Windows 11 / Intel Core i5-4460 **Status:** Resolved

1. Docker Daemon API Connection Failure

Executing standard container interface directives (such as `docker run hello-world`) generated an isolated local communication fault:

```
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
```

Root Cause: The Docker command line binary was operational, but the necessary underlying Windows service layer (`com.docker.service`) was inactive.

2. Windows Service Privilege Escalation

Invoking `Start-Service *docker*` inside standard console layers returned an access constraint:

```
Start-Service : Service 'Docker Desktop Service (com.docker.service)' cannot be started due to the following error: Cannot open com.docker.service service on computer '.'.
```

Resolution: The execution scope was shifted to a dedicated **Administrator: Windows PowerShell** workspace, which effectively allowed full clearance to kickstart the system service pipeline.

3. Hardware Virtualization Barrier (BIOS Level)

Once services were up, Docker triggered a system requirements alert:

```
Virtualization support not detected
Docker Desktop failed to start because virtualisation support wasn’t detected.
```

Root Cause: Task Manager evaluation verified that while the Intel Core i5-4460 processor completely supported virtualization architectures, its status was explicitly set to `Disabled` in the motherboard firmware layout.

4. Outdated Subsystem Environment

Following hardware activation, Docker Desktop required a final dependency update to establish structural stability for its modern container runtimes, prompting a **"WSL needs updating"** blocking frame.

5. Unified Sequential Resolution Playbook

Apply these sequential procedures to resolve the complete stack and bring Docker back to normal operation:

Step 1 Hardware Firmware Acceleration (BIOS)

Restart the computer and access the hardware configuration utilities (typically by holding `Del` or `F2` continuously during boot screens). Locate the CPU configuration blocks and configure the system values exactly as follows:

- **Intel Virtualization Technology / VMX / VT-x:** Toggle to `Enabled`
- Commit the update configuration profiles directly by using the `F10` (Save & Exit) instruction.

Step 2 Initialize Elevated Administrative Workspaces

Search for PowerShell in the system layout, right-click, and select **Run as administrator**. Launch the engine wrapper cleanly:

```
Start-Service *docker*
```

Step 3 Upgrade WSL Virtual Core Components

Synchronize your operating system's local Linux translation layout kernels with official upstream builds by typing:

```
wsl --update
```

Step 4 Cycle Subsystem Layers & Execute Verification

Flush out any stuck hypervisor processes and verify complete stack operational health:

```
wsl --shutdown
docker run hello-world
```